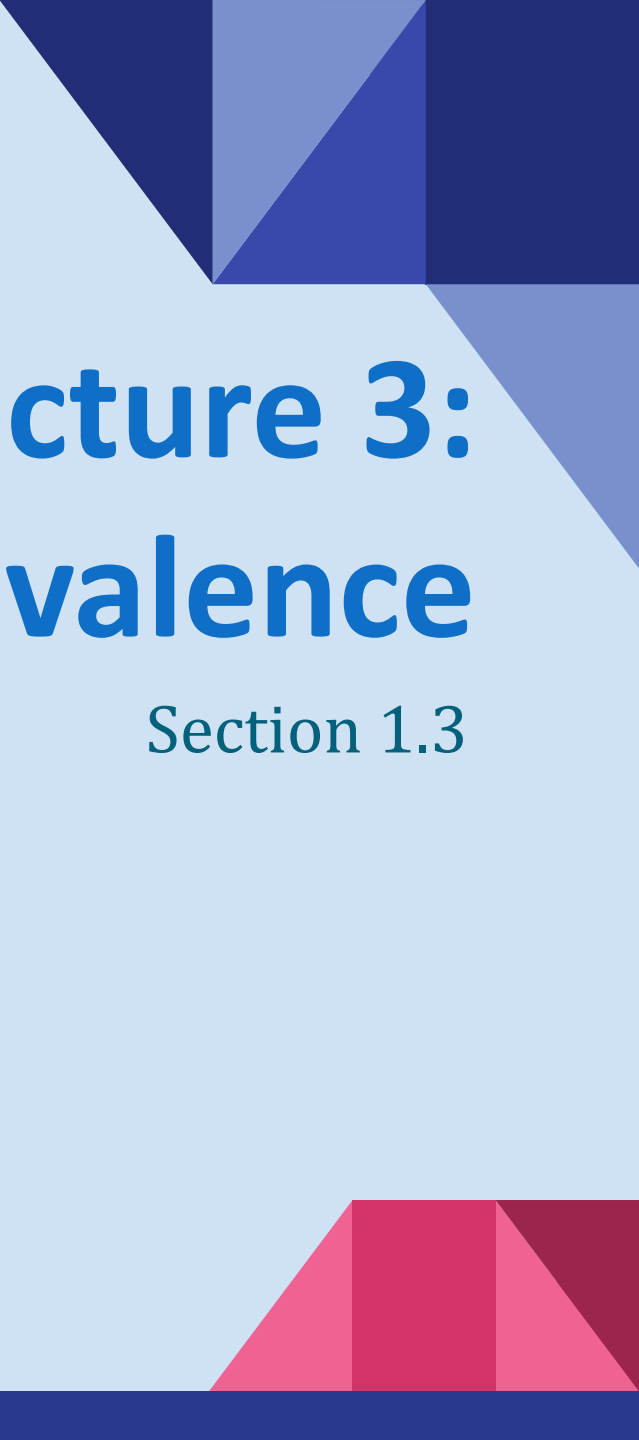


The slide features a light blue background. In the top right corner, there are several overlapping triangles in shades of dark blue and medium blue. In the bottom right corner, there are overlapping triangles in shades of red and pink. The main title is centered and reads:

Lecture 3: Propositional Equivalence

Section 1.3

Tautologies, Contradictions, and Contingencies

- A *tautology* is a proposition which is always true.
 - Example: $p \vee \neg p$
- A *contradiction* is a proposition which is always false.
 - Example: $p \wedge \neg p$
- A *contingency* is a proposition which is neither a tautology nor a contradiction, such as p

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F

Logical Equivalence: Examples

- Two compound propositions p and q are *logically equivalent* if $p \leftrightarrow q$ is a tautology.
- We write this as $p \Leftrightarrow q$ or as $p \equiv q$ where p and q are compound propositions.
- Two compound propositions p and q are equivalent if and only if the columns in a truth table giving their truth values agree.
- This truth table shows that $\neg p \vee q$ is equivalent to $p \rightarrow q$.

p	q	$\neg p$	$\neg p \vee q$	$p \rightarrow q$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

Example: De Morgan's Laws

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg(p \vee q) \equiv \neg p \wedge \neg q$$

Augustus De Morgan

1806-1871



This truth table shows that De Morgan's Second Law holds.

p	q	$\neg p$	$\neg q$	$(p \vee q)$	$\neg(p \vee q)$	$\neg p \wedge \neg q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

Example: Distributive Laws

$$(p \vee (q \wedge r)) \equiv (p \vee q) \wedge (p \vee r)$$

$$(p \wedge (q \vee r)) \equiv (p \wedge q) \vee (p \wedge r)$$

This truth table shows that distributive law of disjunction over conjunction holds.

p	q	r	$q \wedge r$	$p \vee (q \wedge r)$	$p \vee q$	$p \vee r$	$(p \vee q) \wedge (p \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	T	T	T	T
T	F	T	F	T	T	T	T
T	F	F	F	T	T	T	T
F	T	T	T	T	T	T	T
F	T	F	F	F	T	F	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

Predicates and Quantifiers

Section 1.4

Section Summary

- Predicates
- Variables
- Quantifiers
 - Universal Quantifier
 - Existential Quantifier
- Negating Quantifiers
 - De Morgan's Laws for Quantifiers
- Translating English to Logic

Introducing Predicate Logic

- Predicate logic uses the following new features:
 - Variables: x, y, z
 - Predicates: $P(x), M(x)$
 - Quantifiers (*to be covered in a few slides*):
- *Propositional functions* are a generalization of propositions.
 - They contain variables and a predicate, e.g., $P(x)$
 - Variables can be replaced by elements from their ***domain***.

Propositional Functions

- Propositional functions become propositions (and have truth values) when their variables are each replaced by a value from the *domain* (or *bound* by a quantifier, as we will see later).
- For example, let $P(x)$ denote “ $x > 0$ ” and the domain be the integers. Then:
 - $P(-3)$ is false.
 - $P(0)$ is false.
 - $P(3)$ is true.
- Often the domain is denoted by U . So in this example U is the integers.

Examples of Propositional Functions

- Let “ $x + y = z$ ” be denoted by $R(x, y, z)$ and U (for all three variables) be the integers.
Find these truth values:
 - $R(2, -1, 5)$ **Solution: F**
 - $R(3, 4, 7)$ **Solution: T**
 - $R(x, 3, z)$ **Solution: Not a Proposition**
- Now let “ $x - y = z$ ” be denoted by $Q(x, y, z)$, with U as the integers. Find these truth values:
 - $Q(2, -1, 3)$ **Solution: T**
 - $Q(3, 4, 7)$ **Solution: F**
 - $Q(x, 3, z)$ **Solution: Not a Proposition**

Compound Expressions

- If $P(x)$ denotes “ $x > 0$,” find these truth values:
 - $P(3) \vee P(-1)$ **Solution** : T
 - $P(3) \wedge P(-1)$ **Solution** : F
 - $P(3) \rightarrow P(-1)$ **Solution** : F
 - $P(3) \rightarrow \neg P(-1)$ **Solution** : T
- Expressions with variables are not propositions and therefore do not have truth values. For example,
 - $P(3) \wedge P(y)$
 - $P(x) \rightarrow P(y)$
- When used with quantifiers (to be introduced next), these expressions (propositional functions) become propositions.



Charles Peirce (1839-1914)

Quantifiers

- We need *quantifiers* to express the meaning of English words including *all* and *some*:
 - “All men are Mortal.”
 - “Some cats do not have fur.”
- The two most important quantifiers are:
 - *Universal Quantifier*, “For all,” symbol: $\forall$
 - *Existential Quantifier*, “There exists,” symbol: $\exists$
- We write as in $\forall x P(x)$ and $\exists x P(x)$.
- $\forall x P(x)$ means $P(x)$ is true for every x in the *domain*.
- $\exists x P(x)$ means $P(x)$ is true for some x in the *domain*.
- The quantifiers are said to bind the variable x in these expressions.

Universal Quantifier

- $\forall x P(x)$ is read as “For all x , $P(x)$ ” or “For every x , $P(x)$ ”

Examples :

- If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\forall x P(x)$ is false.
- If $P(x)$ denotes “ $x > 0$ ” and U is the positive integers, then $\forall x P(x)$ is true.
- If $P(x)$ denotes “ x is even” and U is the integers, then $\forall x P(x)$ is false.

Existential Quantifier

- $\exists x P(x)$ is read as “For some x , $P(x)$ ”, or as “There is an x such that $P(x)$,” or “For at least one x , $P(x)$.”

Examples :

- If $P(x)$ denotes “ $x > 0$ ” and U is the integers, then $\exists x P(x)$ is true. It is also true if U is the positive integers.
- If $P(x)$ denotes “ $x < 0$ ” and U is the positive integers, then $\exists x P(x)$ is false.
- If $P(x)$ denotes “ x is even” and U is the integers, then $\exists x P(x)$ is true.

Thinking about Quantifiers

- When the domain is finite, we can think of quantification as looping through the elements of the domain.
- To evaluate $\forall x P(x)$ loop through all x in the domain.
 - If at every step $P(x)$ is true, then $\forall x P(x)$ is true.
 - If at a step $P(x)$ is false, then $\forall x P(x)$ is false and the loop terminates.
- To evaluate $\exists x P(x)$ loop through all x in the domain.
 - If at some step, $P(x)$ is true, then $\exists x P(x)$ is true and the loop terminates.
 - If the loop ends without finding an x for which $P(x)$ is true, then $\exists x P(x)$ is false.
- Even if the domains are infinite, we can still think of the quantifiers this fashion, but the loops will not terminate in some cases.

Properties of Quantifiers

- The truth value of $\exists x P(x)$ and $\forall x P(x)$ depend on both the propositional function $P(x)$ and on the domain U .
- **Examples :**
 - If U is the positive integers and $P(x)$ is the statement “ $x < 2$ ”, then **$\exists x P(x)$ is true, but $\forall x P(x)$ is false**.
 - If U is the negative integers and $P(x)$ is the statement “ $x < 2$ ”, then **both $\exists x P(x)$ and $\forall x P(x)$ are true**.
 - If U consists of 3, 4, and 5, and $P(x)$ is the statement “ $x > 2$ ”, then **both $\exists x P(x)$ and $\forall x P(x)$ are true**.
 - But if $P(x)$ is the statement “ $x < 2$ ”, then **both $\exists x P(x)$ and $\forall x P(x)$ are false**.

Precedence of Quantifiers

- The quantifiers $\forall$ and $\exists$ have higher precedence than all the logical operators.
- For example, $\forall x P(x) \vee Q(x)$ means $(\forall x P(x)) \vee Q(x)$
- $\forall x (P(x) \vee Q(x))$ means something different.
- Unfortunately, often people write $\forall x P(x) \vee Q(x)$ when they mean $\forall x (P(x) \vee Q(x))$.

Translating from English to Logic

Example 1 : Translate the following sentence into predicate logic: “Every **student** in this class has taken a course in **Java**.”

Solution :

First decide on the domain U .

Solution 1 : If U is all students in this class, define a propositional function $J(x)$ denoting “ x has taken a course in Java” and translate as $\forall x J(x)$.

Solution 2 : But if U is all people, also define a propositional function $S(x)$ denoting “ x is a student in this class” and translate as $\forall x (S(x) \rightarrow J(x))$.

$\forall x (S(x) \wedge J(x))$ is not correct. What does it mean?

Translating from English to Logic

Example 2 : Translate the following sentence into predicate logic: “Some **student** in this class has taken a course in **Java**.”

Solution :

First decide on the domain U .

Solution 1 : If U is all students in this class, translate as

$$\exists x J(x)$$

Solution 2 : But if U is all people, then translate as

$$\exists x (S(x) \wedge J(x))$$

$\exists x (S(x) \rightarrow J(x))$ is not correct. What does it mean?

Equivalences in Predicate Logic

- Statements involving predicates and quantifiers are *logically equivalent* if and only if they have the same truth value
 - for every predicate substituted into these statements and
 - for every domain of discourse used for the variables in the expressions.
- The notation $S \equiv T$ indicates that S and T are logically equivalent.
- **Example :** $\forall x \neg \neg S(x) \equiv \forall x S(x)$

Thinking about Quantifiers as Conjunctions and Disjunctions

- If the domain is finite, a universally quantified proposition is equivalent to a conjunction of propositions without quantifiers and an existentially quantified proposition is equivalent to a disjunction of propositions without quantifiers.
- If U consists of the integers 1,2, and 3:

$$\forall x P(x) \equiv P(1) \wedge P(2) \wedge P(3)$$

$$\exists x P(x) \equiv P(1) \vee P(2) \vee P(3)$$

- Even if the domains are infinite, you can still think of the quantifiers in this fashion, but the equivalent expressions without quantifiers will be infinitely long.

Negating Quantified Expressions

- Consider $\forall x J(x)$

“Every student in your class has taken a course in Java.”

Here $J(x)$ is “x has taken a course in Java” and the domain is students in your class.

- Negating the original statement gives “*It is not the case that every student in your class has taken Java.*” This implies that “There is a student in your class who has not taken Java.”

Symbolically $\neg \forall x J(x)$ and $\exists x \neg J(x)$ are equivalent

Negating Quantified Expressions (continued)

- Now Consider $\exists x J(x)$

“There is a student in this class who has taken a course in Java.”

Where $J(x)$ is “x has taken a course in Java.”

- Negating the original statement gives “It is not the case that there is a student in this class who has taken Java.” This implies that “Every student in this class has not taken Java”

Symbolically $\neg \exists x J(x)$ and $\forall x \neg J(x)$ are equivalent

De Morgan's Laws for Quantifiers

- The rules for negating quantifiers are:

TABLE 2 De Morgan's Laws for Quantifiers.			
<i>Negation</i>	<i>Equivalent Statement</i>	<i>When Is Negation True?</i>	<i>When False?</i>
$\neg \exists x P(x)$	$\forall x \neg P(x)$	For every x , $P(x)$ is false.	There is an x for which $P(x)$ is true.
$\neg \forall x P(x)$	$\exists x \neg P(x)$	There is an x for which $P(x)$ is false.	$P(x)$ is true for every x .

- The reasoning in the table shows that:

$$\neg \forall x P(x) \equiv \exists x \neg P(x)$$

$$\neg \exists x P(x) \equiv \forall x \neg P(x)$$

Translation from English to Logic

Examples :

1. “Some student in this class has visited Mexico.”

Solution : Let $M(x)$ denote “x has visited Mexico” and $S(x)$ denote “x is a student in this class,” and U be all people.

$$\exists x (S(x) \wedge M(x))$$

2. “Every student in this class has visited Canada or Mexico.”

Solution : Add $C(x)$ denoting “x has visited Canada.”

$$\forall x (S(x) \rightarrow (M(x) \vee C(x)))$$

Some Translating from English into Logical Expressions

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

Translate “Everything is fast.”

Solution: $\forall x F(x)$

Translation (cont)

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

“Nothing is slow.”

Solution: $\neg \exists x S(x)$ What is this equivalent to?

Solution: $\forall x \neg S(x)$

Translation (cont)

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

“All fast things are slow.”

Solution: $\forall x (F(x) \rightarrow S(x))$

Equivalent to: For all things, fast implies slow.

Translation (cont)

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

“Some fast things are turning.”

Solution: $\exists x (F(x) \wedge T(x))$

Equivalent to: Some things are fast and turning

Translation (cont)

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

“No slow thing is turning.”

Solution: $\neg \exists x (S(x) \wedge T(x))$ What is this equivalent to?

Solution: $\forall x (\neg S(x) \vee \neg T(x))$

Translation (cont)

- $U = \{\text{fast, slow, turning}\}$

$F(x)$: x is fast

$S(x)$: x is slow

$T(x)$: x is turning

“If anything fast is a slow then it is also a turning.”

Solution: $\forall x ((F(x) \wedge S(x)) \rightarrow T(x))$



Charles Lutwidge Dodgson
(AKA Lewis Carroll)
(1832-1898)

Lewis Carroll Example

- The first two are called *premises* and the third is called the *conclusion*.
 - “All lions are fierce.”
 - “Some lions do not drink coffee.”
 - “Some fierce creatures do not drink coffee.”
- Here is one way to translate these statements to predicate logic. Let $P(x)$, $Q(x)$, and $R(x)$ be the propositional functions “ x is a lion,” “ x is fierce,” and “ x drinks coffee,” respectively.
 - $\forall x (P(x) \rightarrow Q(x))$
 - $\exists x (P(x) \wedge \neg R(x))$
 - $\exists x (Q(x) \wedge \neg R(x))$
- Later we will see how to prove that the conclusion follows from the premises.